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From Insights to Impact: Using Data to Drive Diversity and Organizational Excellence

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Abstract

The objective of this paper is to explore the application of data-driven strategies in promoting organizational excellence by addressing critical organizational insights. In particular, with a focus on using data to inform and monitor diversity and inclusion targets to drive people development and diversity initiatives in the energy industry.

Aramco is launching an innovative Integrated Organization Intelligence Tool (iOIT) in 2025. It features a unified view of organizations and strategic support. Built for the specific needs of Aramco's Organization Consulting Professionals, the tool offers a suite of features, including Organizational Evolvment and Growth, which visualizes the organizational structure and changes over time. Additionally, Job Utilization and Workforce Analysis to deliver valuable insights on workforce demographics and optimization opportunities, providing a holistic view of the organization's workforce and identifying areas of imbalance and track the effectiveness of interventions aimed at promoting diversity and inclusion.

The iOIT is expected to provide a data-driven platform to unlock organizational excellence and promote diversity and inclusion, by addressing biases in hiring, promotion practices and tracking diversity and inclusion KPIs, enabling targeted interventions that address specific gaps and challenges. By analyzing workforce data, organizations can identify areas where diversity and inclusion initiatives are needed most, and track the impact of these initiatives over time. Critical lessons learned from the application of the iOIT include the importance of using data to inform diversity and inclusion strategies, and the need for continuous monitoring and evaluation to ensure the effectiveness of these initiatives.

This paper presents novel information on the application of data-driven strategies in promoting diversity and inclusion, utilizing not only workforce data but also jobs and organizations to make informed targeted interventions that drive people development and diversity initiatives. The iOIT's innovative approach empowers organizations to make data-driven decisions that promote a more diverse and inclusive workplace, ultimately contributing to the industry's overall growth and success.

Introduction

Data analysis has been relatively straightforward when it comes to for example, analyzing an organization performance at the departmental level, where simple metrics such as headcount and number of vacant chief

position were sufficient to highlight an organization pain area. However, as the organization grew the need for more nuanced insights arose, the complexity of analysis increased, particularly when incorporating additional data points from various sources, such as HR systems, financial reports, and operational metrics.

To address this challenge, Aramco is introducing the Integrated Organization Intelligence Tool (iOIT), a multi-faceted platform designed to monitor key performance indicators (KPIs) and organizational health from various perspectives, providing a holistic view of the organization's performance. The iOIT enhances the evaluation of organizational excellence, which refers to the organization's ability to achieve its strategic objectives, and organizational health, which encompasses the well-being and productivity of its workforce, across three primary dimensions: organization evolution, job utilization, and workforce insights. For the organization section, users can monitor headcount, track evolution, and easily report on organization KPIs including the number of created, deleted, and updated organizations. For job utilization, users can analyze and monitor their job authorization and utilization, drawing insights to check job attrition and unplaced job numbers, and build on job application requests. As for HR business partners, they can also utilize this section to monitor their jobs authorized to other organizations in the company and track their utilization. Finally, for the workforce section, which this paper focuses on to draw insight about the diversity and retention metrics of the company's workforce, users can optimize their workforce planning by understanding their workforce trends and attrition curves. Ultimately, supporting a data-driven approach in decision making for workforce planning and management. By providing this holistic view we drive transparent, data-driven insights that enables collaborative decision-making with the company's consultant and HR business partner, ensuring that organizations are built for clarity, agility, and value.

The iOIT tool is powered by QlikSense, a business intelligence platform that enables interactive and intuitive data visualization, alongside Vizlib, a library of advanced visualization extensions that enhance the capabilities of QlikSense and a Business Explorer (Bex) Query designer to access data stored in the company's data warehouse. Together, these tools provide a 3-dimensional view of organizational health, allowing for assessment, improvement, and sustainability of organizational performance and its people. This integrated approach overcomes traditional challenges of data silos, manual reporting, and limited visibility, providing a unified platform for data-driven decision-making.

This paper outlines the methodology employed to analyze organizational structure and evolution, workforce demographics, and job utilization, with a focus on identifying strategic opportunities for growth and optimization. By integrating these three dimensions, we can gain a deeper understanding of the complex relationships between them and inform data-driven decisions that drive diversity and organizational excellence. The approach taken emphasizes a holistic analysis of the organization, leveraging a nuanced understanding of the interplay between structure, demographics, and job utilization to uncover areas for improvement and optimization.

Statement of Theory and Definitions

The iOIT tool measures and assesses the health metrics of each of the three aspects, the Organization, Job Design and workforce Health-checks. This framework utilizes a range of metrics to assess organizational efficiency and workforce effectiveness. One standard metric usually looked at to assess an organization health is the Span of Control, which measures the number of organizational entities reporting to a managerial position, with healthy ranges varying by role type. For example, if a manager has only 2 direct reports, it may suggest that they are underutilized and that the organization's structure is inefficient and vice versa, if the manager has many direct reports it may indicate that they are overburdened and unable to provide adequate guidance and support, potentially leading to decreased productivity and employee satisfaction. Additionally, monitoring average number of individuals per organization hierarchy can help you identify under/over populated organizations that are below/above the average headcount of that organization hierarchy. Moving on to the next section which is job utilization, the tool can identify any outdated jobs, which tracks the

number of jobs owned by different organizations that have not been updated for a couple of years, it's important to track the job evaluation status and flag any 'due to evaluation' jobs to avoid outdated jobs. Users can also monitor job authorized by the different organization and measure the growth of jobs created per year. Finally, Workforce Metrics, on the other hand, include variant demographics users can view the age distribution and determine the seniority of an organization population, by measuring the percentage of employees at age 35 and above. Users can also monitor the percentage of Top Performers Voluntary Attrition, which assesses the percentage of top performers who left the company over total voluntary attrition. Looking at the number of Permanent Employees and Chief Position Holders (CPH) in one organization, which measures the percentage of employees permanently assigned to a given organization, can give you insights regarding the stability of that organization. While it's important to measure such metrics it's important to highlight to take into consideration that every organization is unique, thus establishing a healthy range of such metrics should take into consideration the dynamic of an organization. The company have established those ranges based on its internal standard framework.

Description and Application of Equipment and Processes

Several technologies are used in the development of a comprehensive front-end solution to address workforce reporting, scenarios planning, and predictive analysis which is a pivotal aspect of this project. For example, an in-house developed solution for Workforce Insights (HRISC) serves as a prime example of Qlik efficacy. This innovative tool has been designed to provide actionable insights, empowering users to develop multiple scenarios using input variables, thereby facilitating informed decision-making. To further augment the analysis capabilities of the HRISC, we have leveraged additional Qlik Sense extensions, such as Heatmaps and Sankey charts, which enable the visualization of complex data relationships and patterns. These advanced visualization tools have significantly enhanced the dashboard's ability to convey nuanced insights, allowing users to identify trends, spot anomalies, and forecast future outcomes with greater accuracy. Additionally, the development process of multiple tools, including Contracts, Medical Leave, and Employee Performance, has been guided by an in-house developed methodology that incorporates an iterative process. This methodology involves iteratively refining the back-end model, enhancing front-end queries using set-analysis, and fine-tuning the dashboard's overall performance. Fig.1 demonstrates the development methodology process

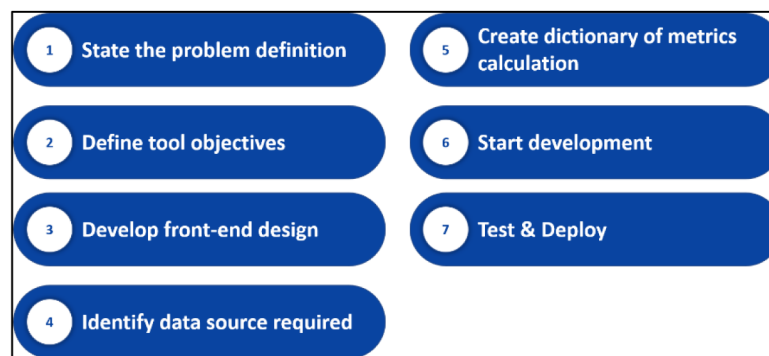


Figure 1—In-house development methodology

By adopting this iterative approach, we have been able to ensure that each dashboard is tailored to meet the specific needs of its intended users, providing them with relevant, timely, and actionable information. The same methodology has been successfully applied to govern the overall dashboards development process, with notable examples including the Supplemental Manpower (SMP) onboarding and Professional Development Program (PDP) Analysis dashboards. To further enhance the front-end chart functionality, we have utilized Vizlib, a powerful tool that provides greater control over chart customization by allowing

us to write CSS. Vizlib's extensive library of customized charts has enabled us to create embedded charts that provide a second layer of information simultaneously, thereby offering enhanced analytics capabilities. Fig.2 demonstrates an embedded chart from Vizlib

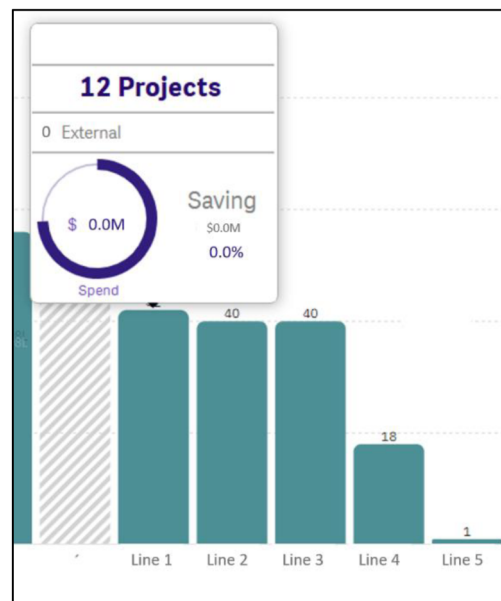


Figure 2—Embedded Chart using Vizlib extension

In terms of data connectivity, we have established seamless integration between Qlik Sense and various data sources using connectors such as REST API (a standardized interface for interacting with web services), which enables us to fetch data from SAP databases using SAP BEx Queries (a reporting tool used in SAP Business Intelligence (BI) systems which allows users to create queries). We have also connected to MS SharePoint (a collaborative platform for document management and content sharing), and utilized OLEDB/ODBC connectors (standardized interfaces for accessing relational databases) to integrate data from traditional databases like MS SQL Server and Oracle, ensuring that all relevant data is accessible and up-to-date. To automate the process of updating dashboards, we have leveraged Qlik Sense's Task scheduler, which enables us to schedule tasks to run at regular intervals, providing users with the most current information across all dashboards. This automation has not only streamlined our workflow but also ensured that our dashboards remain relevant, accurate, and reliable, thereby supporting informed decision-making and driving business success. By combining these technologies and methodologies, we have created a robust and scalable front-end solution that addresses the complex needs of workforce reporting, scenarios planning, and predictive analysis, ultimately empowering organizations to make data-driven decisions and drive strategic growth. The iOIT features a comprehensive Workforce Analytics and Insights platform, designed to provide a deeper understanding of the company's workforce. This platform tracks key metrics such as:

- **Workforce Growth:** Monitor the overall growth of our workforce, including new hires, promotions, and internal mobility.
- **Internal Mobility:** Track employee movements within the organization, including transfers, promotions, and role changes.
- **Workforce Development:** Analyze employee training, development, and upskilling initiatives to ensure our workforce is equipped with the necessary skills to drive business success.
- **Mandatory Learning:** Monitor compliance with mandatory training and development programs, ensuring our workforce is up-to-date with industry regulations and best practices.

- Diversity and Inclusion: Track diversity metrics, including gender, age, ethnicity, and other demographic characteristics, to ensure our workforce reflects the communities we serve.
- Age and Job Distribution: Analyze the age distribution of our workforce and job categories to identify potential skills gaps and succession planning opportunities.
- Time Management: Monitor overtime, medical time, and personal time to ensure our workforce is managed efficiently and effectively.
- Hiring Efforts: Analyze hiring trends, including diversity and inclusion metrics, to identify areas for improvement and optimize our recruitment strategies.
- Attrition vs. Hiring and Workforce Growth: Compare attrition rates to hiring and workforce growth to identify potential issues and opportunities for improvement.

Additionally, the platform monitors diversity and inclusion hiring metrics to track and enhance our hiring practices, these metrics includes:

- Organization-wise Hiring: Identify which organizations are hiring the most diverse talent and replicate their strategies.
- Job-wise Hiring: Analyze which jobs are most in demand and develop strategies to attract diverse candidates for those roles.
- Gender and Age Distribution: Monitor the gender and age distribution of new hires to ensure our workforce reflects the diversity of the communities we serve.
- Build vs. Buy: Determine whether we are building or buying talent for key roles and develop strategies to build a strong pipeline of internal candidates.

To reduce attrition and improve workforce retention, the platform also conducts an attrition analysis, examining the following areas:

- Organization-wise Attrition: Identify which organizations are experiencing high attrition rates and develop targeted retention strategies.
- Job-wise Attrition: Analyze which jobs are most prone to attrition and develop strategies to improve job satisfaction and engagement.
- GC, Location, and Years of Service: Examine the impact of geographic location, years of service, and other factors on attrition rates.
- Attrition Reasons: Identify the root causes of attrition, including performance, lack of opportunities, or other factors, and develop strategies to address these issues.
- Performance and Attrition: Analyze the relationship between employee performance and attrition to identify opportunities to improve retention and development strategies.

Presentation of Data and Results

The iOIT tool has been designed to integrate three main sections. The first section is about the history of the organization structure in Saudi Aramco captures the evolution of Saudi Aramco's organizational structure over time, including growth of organization, merging and restructuring efforts. It provides valuable context to enable users to understand how organization has adapted to changing market conditions and business needs. The second section is about Job descriptions, contains a repository of job descriptions for various roles within Saudi Aramco. This section provides detailed information on job responsibilities, requirements, and competencies, which gives users clarity in role expectations. The final section is about the Saudi Aramco workforce including employee headcount, distribution, and movement across different organizational levels, departments, and locations. This section provides insights into workforce trends and more. It enables users to make data-driven decision and monitor diversity and inclusion metrics. The landing page of this tool is visualized in [Fig.3](#).

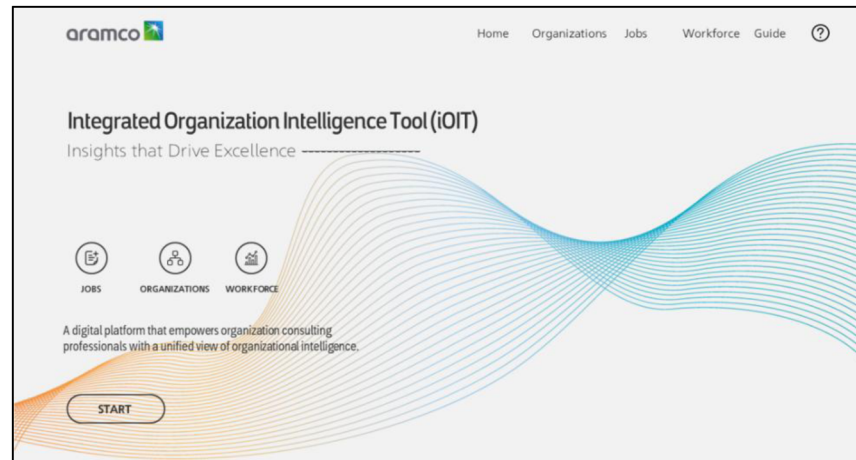


Figure 3—Snapshot of iOIT landing page

Section 1: Organization evolvement

Starting with the first section, the user input two date points, we name them as a previous date and current date. Once those two date points are entered, the tool works behind the scene to come up with the difference in the organization structure between these dates. Fig.4 shows date filters and run button to run the analysis.

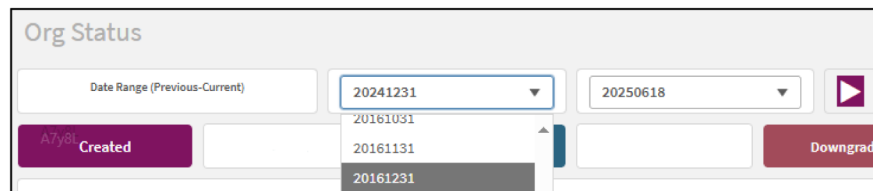


Figure 4—Snapshot of Date Range Selection

This allows to track and analyzes the Organizations Hierarchy level differences for example, number of departments, divisions, groups etc. and compares how many existed back then in comparison to the current date. The user will also be able to see the following metrics of Organization Evolvement:

- Number of organizations created
- Number of organizations deleted
- Number of organizations re-titled
- Number of organizations upgraded
- Number of organizations downgraded

A way to visualize this change is to utilize Sankey chart provided from Vizlib, for example Fig.5 shows the flow of upgraded organizations.

The user will be able to see full details and the raw data of all of these changes. The results are crucial for organization analysts and management to gauge and supervise organization changes and ensure the changes meet the Company growth and strategic objectives.

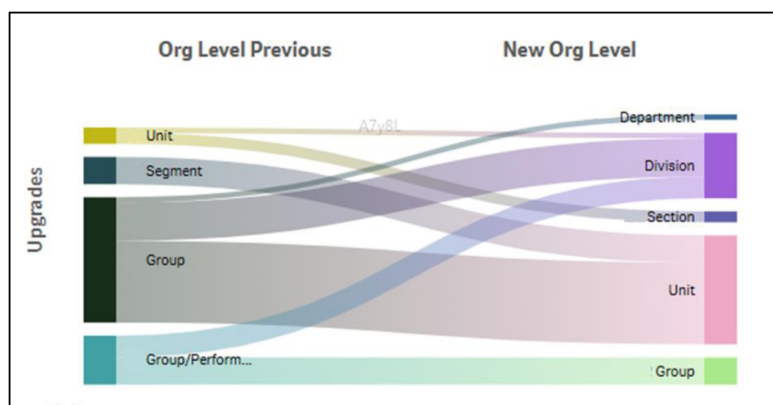


Figure 5—Vizlib Sankey graph of upgraded organizations

Section 2: Job Utilization

The second section provides a comprehensive information about jobs within Saudi Aramco. In terms of job hierarchy composition, this starts with a Job Family Category (Industrial, Professional, and Administrative) to Job Family Groups, then, finally to the Job Title (the job that employee hold in a specific Grade Code). This section in the dashboard is informative and provide critical information for organization analyst to utilize jobs within specific organization. The section provides detailed information about jobs like job description, general tasks duties, responsibilities and job requirements. The importance of this information cannot be overstated, given that the jobs in Saudi Aramco are diverse unlike other oil Companies; and analyst do require information about these jobs to ensure that jobs are utilized in the right organization. For instance, you cannot have an oil/gas operator work in the Information Technology (IT) department as the job is not authorized to be used in IT department. Where in fact, you will expect this job to be authorized in the oil plants organization. This is only one example to explain the authorization aspects. However, what makes Saudi Aramco unique, as an oil company, that it has jobs in so many aspects like security, education, training, development, IT, recruitment, pilots, and the usual oil related jobs. The diversity of jobs within the company is a significant strength, offering a wide range of opportunities and expertise. To harness this potential, this section provides a clear and comprehensive framework, empowering organizational analysts to make informed decisions and optimize job utilization. For example, users can utilize the section to look at the job utilization trend historically and highlight under-utilized jobs. Fig.6 shows dummy data of job utilization since 2012.

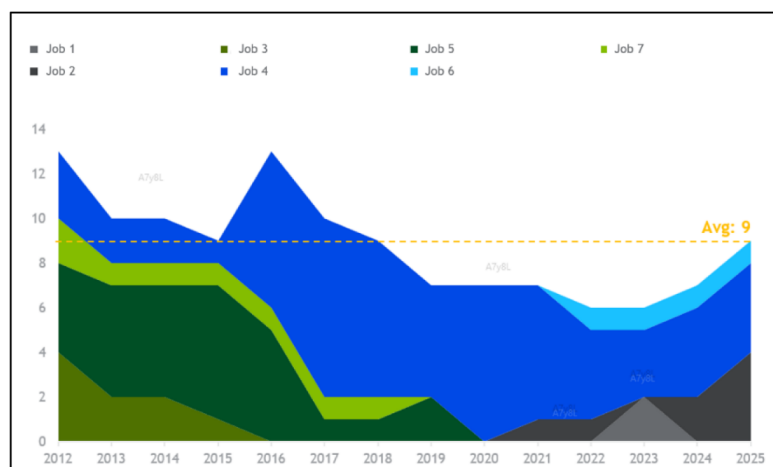


Figure 6—Job utilization since 2012

By doing so, we can ensure that each job is utilized to its fullest potential, maximizing efficiency and effectiveness across the organizations.

Section 3: Workforce Insights

The third section is about the Saudi Aramco workforce, the workforce profile provides a comprehensive overview of an organization's workforce, it has trends and analysis where HR business partner utilize to share key insights with management for decision making and proactive planning. Such demographic included in this section is as follow

- Demographic data: age, gender, nationality, education level, and job function
- Workforce structure: Internal Labor Market tree, organizational hierarchy workforce headcount, departmental breakdown, and job categories
- Diversity and inclusion metrics: representation of underrepresented groups, diversity ratios, and inclusion in terms of gender, disability, and female in managerial positions
- Workforce Movements: Hiring, attrition, turnover rates, retention rates, and tracking top performer

One of the key strengths of this workforce profile is its ability to provide such information and insights at any organization level in Saudi Aramco. The user also has the ability to filter by gender, employee type, job title, age category, and any other available fields for selections in the tool. Fig.7 is a snapshot of a small segment from the workforce section.

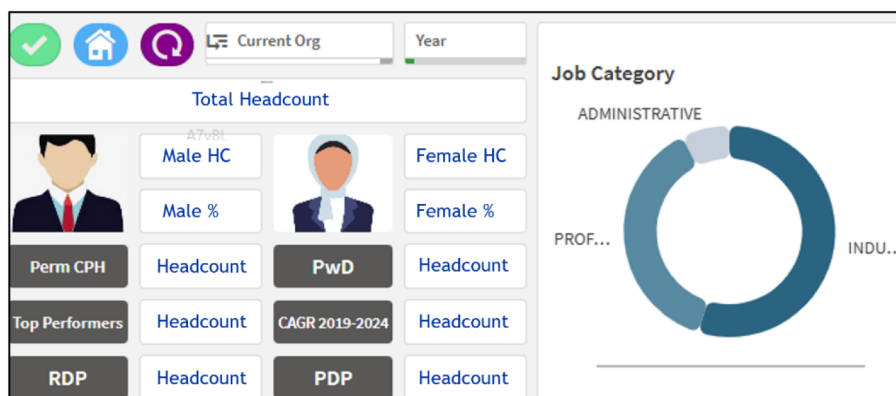


Figure 7—Snippet of Workforce Metrics

All of these features happen interactively and instantly and is a significant improvement over traditional methods, which required manual data extraction and metric development. With this tool, users can quickly and easily access the information they need to make informed decisions and drive business outcomes.

Conclusions

In conclusion, the development and launch of the Integrated Organization Intelligence Tool (iOIT) is essential which will also help towards the goal of maintaining and promoting diversity and inclusion target. By harnessing the power of data analytics, the iOIT provides a comprehensive framework for assessing organizational health, workforce demographics, and job utilization, enabling informed decision-making and proactive planning. The tool's ability to track key metrics such as diversity and inclusion, workforce growth, and internal mobility has the potential to drive business outcomes and promote organizational excellence. The findings of this research highlight the importance of using data to inform diversity and inclusion strategies and the need for continuous monitoring and evaluation. Ultimately, the iOIT has the potential to inform and shape diversity and inclusion policies and practices in the energy industry, contributing to a

more agile, responsive, and effective organization that reflects the diversity of the communities it serves. By leveraging the insights and capabilities of the iOIT, we can use data driven insights to drive meaningful change and promote a culture of inclusion and excellence.